Bio-Inspired Heat Exchanger — Design Summary

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Generated: 2026-06-17 01:46

encapsulation:
  length_mm......................... 15.0
  width_mm.......................... 15.0
  height_mm......................... 20.0
  wall_mm........................... 3.0
  Dh_mm............................. 1.092
  V_int_cm3......................... 1.134
flow:
  v_in_m_s.......................... 0.1
  T_in_K............................ 380.0
  T_ext_K........................... 270.0
  p_out_Pa.......................... 0.0
  Re................................ 109.2
  Pr................................ 6.852
fluid_properties:
  kinematic_viscosity_m2_s.......... 1e-06
  density_kg_m3..................... 1000.0
  conductivity_W_mK................. 0.61
  specific_heat_J_kgK............... 4180.0
  dynamic_viscosity_Pas............. 0.001
  thermal_diffusivity_m2_s.......... 1.459e-07
material_properties:
  solid_conductivity_W_mK........... 237.0
  fluid_conductivity_W_mK........... 0.61
  solid_density_g_mm3............... 0.0027
  rhoc_J_m3K........................ 4180000.0
  darcy_number...................... 1e-09
  hconv_W_m2K....................... 1000.0
  qu_regularisation................. 0.005
sizing:
  Re................................ 1000.0
  Nu................................ 13.178
  wall_thickness_est_mm............. 0.1962
  h_W_m2K........................... 22284.82
lattice:
  tpms.............................. gyroid
  k0_rad_m.......................... 3490.659
  uc_mm............................. 1.8
  phi............................... 0.5646
  iso_level......................... 0.3
  wall_mm........................... 0.2
  unit_mm........................... 1.8
  epsilon_mm........................ 0.2
  kbound_rad_mm..................... 3.4
preview_geometry:
  solidity.......................... 0.1913
  solid_volume_cm3.................. 0.217
  mass_g............................ 0.96
  surface_cm2....................... 39.8
  n_triangles....................... 217864
manufacturability:
  overhang_penalty.................. 0.1075
  thickness_penalty................. 0.4518
  combined_penalty.................. 0.5594
  est_wall_thickness_mm............. 0.1096
  min_feature_size_mm............... 0.2
  max_overhang_angle_deg............ 45.0
  printable......................... False
optimization_settings:
  mode.............................. pressure
  method............................ MMA
  max_iterations.................... 10.0
  parallel_cores.................... 10
  meantT_max_K...................... 303.0
  dissPower_max_W................... 9800.0
  spacing_mm........................ 7.0
  bake_spacing_mm................... 1.4
  pareto_enabled.................... False
am_settings:
  no_overhang....................... False
  am_theta_deg...................... 45.0
  am_L_bridge_mm.................... 1.5
  align_build_to_flow............... True
  build_direction................... flow-aligned
performance:
  mass_flow_rate_kg_s............... 0.0002
  outflow_temperature_K............. 207.8
  delta_T_K......................... -172.2
  heat_transfer_coeff_W_m2K......... 1000.0
  nusselt_number_sim................ 1.79
  nusselt_number_empirical.......... 34.189
  friction_factor_empirical......... 1.06184
  pressure_drop_empirical_Pa........ 336.936
  dissipation_power_empirical_W..... 0.06739
  heat_transfer_empirical_W_m2K..... 19103.306
optimization_results:
  baseline_dissipation_W............ 1355.07
  baseline_mean_temp_K.............. 293.9
  opt_dissipation_W................. 1236.29
  opt_mean_temp_K................... 297.5
  dissipation_reduction_pct......... 8.8
  mean_temp_reduction_pct........... -1.2
lpbf_build_analysis:
  status............................ skipped
  reason............................ PySLM not installed (pip install PythonSLM)
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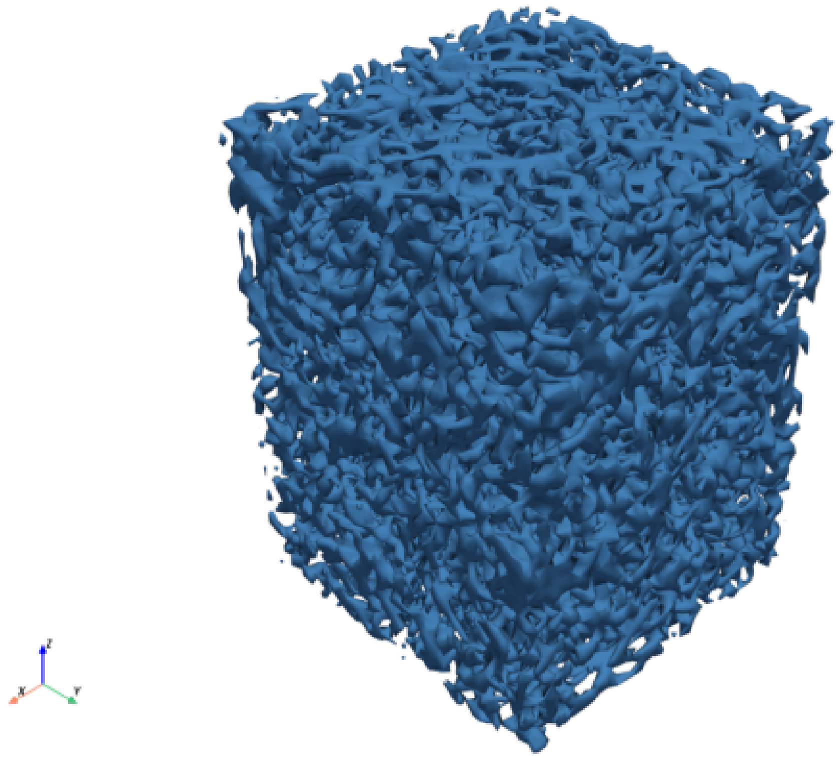
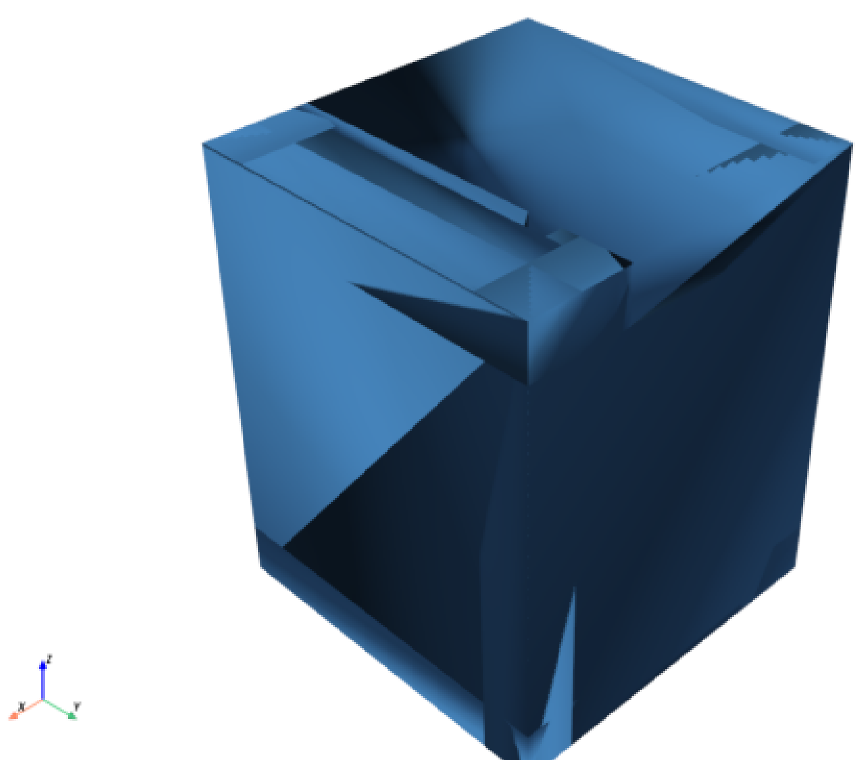
Optimised Geometry

gyroid_surface_lattice.stl

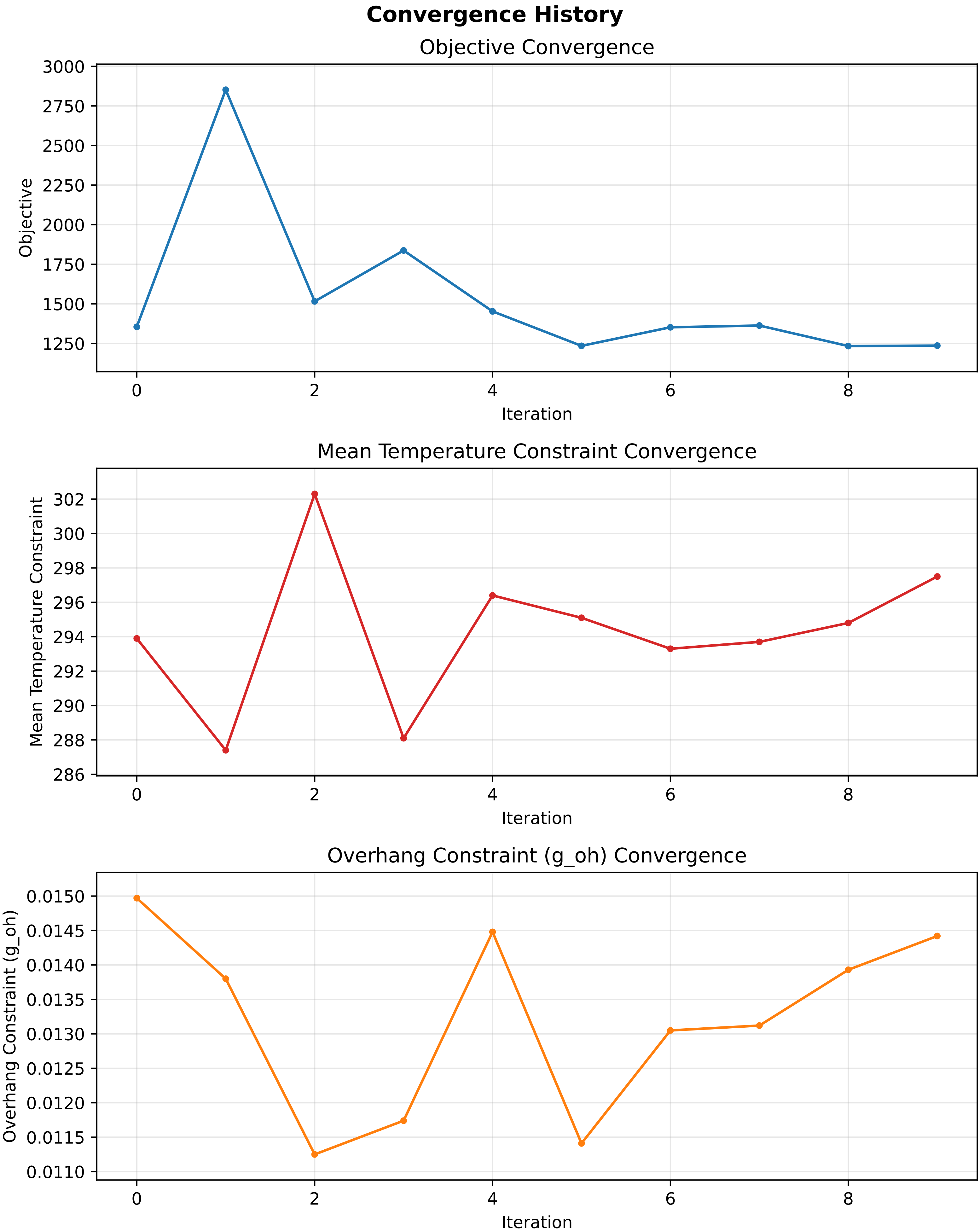


gyroid_surface_encap.stl



Optimisation History

iter	mode	J_meanT	J_obj	J_aug	DissPower	outletT	solid_frac	solid_g	flow_m3s	deltaP_m2s2	wall_thickness	g_meanT	pen_meanT	g_disspower	pen_disspower	mu_penalty	g_oh	pen_oh	mu_oh	grad_norm	delta_x	elapsed_s	alphaMax
1		23.87	1355	1356	1355	281.6	0.4005	4.871	2e-07	6.797	0.2	293.9	0	nan	0	1e+04	0.01497	1.147	8100	1436	nan	296.8	4.444e+07
2		17.38	2852	2853	2852	278.4	0.4001	4.867	2e-07	14.29	0.2	287.4	0	nan	0	1e+04	0.0138	1.202	9720	801.3	18.84	301.7	4.444e+07
3		32.31	1516	1517	1516	286.2	0.401	4.878	2e-07	7.607	0.2	302.3	0	nan	0	1e+04	0.01125	1.126	1.166e+04	1124	18.84	310.2	4.444e+07
4		18.13	1837	1839	1837	279.6	0.4011	4.88	2e-07	9.218	0.2	288.1	0	nan	0	1e+04	0.01174	1.427	1.4e+04	333.5	17.75	303.2	4.444e+07
5		26.39	1453	1455	1453	282.8	0.4029	4.901	2e-07	7.29	0.2	296.4	0	nan	0	1e+04	0.01448	2.535	1.68e+04	1180	17.84	303.2	4.444e+07
6		25.09	1235	1237	1235	282.1	0.3994	4.858	2e-07	6.203	0.2	295.1	0	nan	0	1e+04	0.01141	1.858	2.016e+04	443.5	16.55	302.7	4.444e+07
7		23.27	1352	1355	1352	280.9	0.4001	4.867	2e-07	6.797	0.2	293.3	0	nan	0	1e+04	0.01305	2.806	2.419e+04	623.8	15.95	307.8	4.444e+07
8		23.72	1363	1367	1363	281.6	0.404	4.914	2e-07	6.846	0.2	293.7	0	nan	0	1e+04	0.01312	3.458	2.902e+04	782.1	14.82	302	4.444e+07
9		24.83	1233	1238	1233	282.4	0.402	4.89	2e-07	6.193	0.2	294.8	0	nan	0	1e+04	0.01393	4.398	3.483e+04	295.7	13.79	300.7	4.444e+07
10		27.51	1236	1242	1236	283.6	0.3989	4.852	2e-07	6.211	0.2	297.5	0	nan	0	1e+04	0.01442	5.591	4.179e+04	324.6	13.77	305	4.444e+07



Convergence History
Objective Convergence
Objective
3000
2750
2500
2250
2000
1750
1500
1250
0
2
4
6
8
Iteration
Mean Temperature Constraint Convergence
Mean Temperature Constraint
302
300
298
296
294
292
290
288
286
0
2
4
6
8
Iteration
Overhang Constraint (g_oh) Convergence
Overhang Constraint (g_oh)
0.0150
0.0145
0.0140
0.0135
0.0130
0.0125
0.0120
0.0115
0.0110
0
2
4
6
8
Iteration